

## Topic quiz: Work and energy

IBDP Physics SL · Grade 11 · A.3 Work and energy · 10 multiple-choice questions

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Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

**How to use.** Practise with the simulations and worksheets first. Then answer every question: circle one letter and write it in the grid below. Use  $g = 9.81 \text{ m s}^{-2}$  and show any working next to the question.

**Check yourself online:** the same quiz with instant feedback and explanations — <https://physics.taneski.com/quiz-work-energy/>

[illegible]

### Question 1

A child pulls a sled 10.0 m along flat snow with a force of 50.0 N at  $30^\circ$  above the horizontal. How much work does the pulling force do?

- (A) 43.3 J                      (B) 250 J  
(C) 433 J                      (D) 500 J

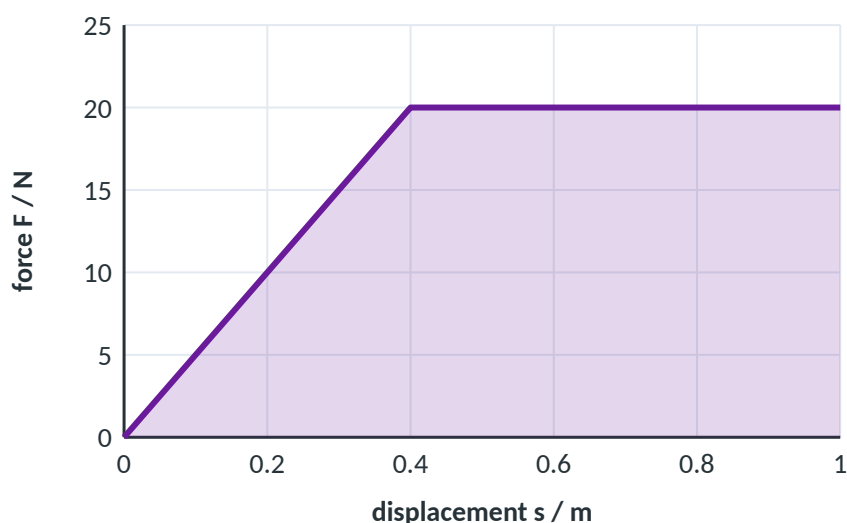
## Question 2

A box is pulled 5.0 m along a horizontal floor. How much work does the normal force do on the box?

- ☐ (A) Zero, because the normal force is perpendicular to the displacement  
☐ (B) Zero, because the normal force is cancelled by the weight  
☐ (C)  $mg \times 5.0 \text{ m}$   
☐ (D) A negative amount, because it opposes the motion

### Question 3

The graph shows the force on a trolley against its displacement. How much work is done over the first 1.0 m?



- |                                |                                |
|--------------------------------|--------------------------------|
| <input type="radio"/> (A) 4 J  | <input type="radio"/> (B) 12 J |
| <input type="radio"/> (C) 16 J | <input type="radio"/> (D) 20 J |

### Question 4

A 20 kg box is raised 1.5 m, either by lifting it straight up or by pushing it up a 3.0 m frictionless ramp at constant speed. Which statement is correct?

- ☐ (A) The ramp needs twice the work, because the distance is twice as long
- ☐ (B) The ramp needs half the work, because the force is half as big
- ☐ (C) Lifting needs no work, because the box ends at rest
- ☐ (D) Both need the same work, 294 J, but the ramp needs a smaller force

### Question 5

A car brakes from  $20 \text{ m s}^{-1}$  and stops in 25 m. With the same braking force, what is its stopping distance from  $40 \text{ m s}^{-1}$ ?

- |                                 |                                 |
|---------------------------------|---------------------------------|
| <input type="radio"/> (A) 25 m  | <input type="radio"/> (B) 50 m  |
| <input type="radio"/> (C) 100 m | <input type="radio"/> (D) 200 m |

### Question 6

A ball is dropped from a height of 5.0 m. Air resistance is negligible. How fast is it moving just before it hits the ground?

- |                                                  |                                                  |
|--------------------------------------------------|--------------------------------------------------|
| <input type="radio"/> (A) $7.0 \text{ m s}^{-1}$ | <input type="radio"/> (B) $9.9 \text{ m s}^{-1}$ |
| <input type="radio"/> (C) $49 \text{ m s}^{-1}$  | <input type="radio"/> (D) $98 \text{ m s}^{-1}$  |

### Question 7

A 0.500 kg Smart Cart rolls down a ramp. It loses 0.513 J of gravitational potential energy and gains 0.483 J of kinetic energy. How much energy is transferred to thermal energy by friction?

- (A) 0 J, because energy is conserved
- (B) 0.030 J
- (C) 0.483 J
- (D) 0.996 J

### Question 8

A heavy pendulum is released from rest touching a student's nose. The student keeps perfectly still. Why can the pendulum not hit the student when it swings back?

- (A) It cannot rise higher than where it started, because its total energy cannot increase
- (B) It loses all its energy at the far end of the swing
- (C) Its mass is too large for it to come back
- (D) Gravity is weaker on the way back

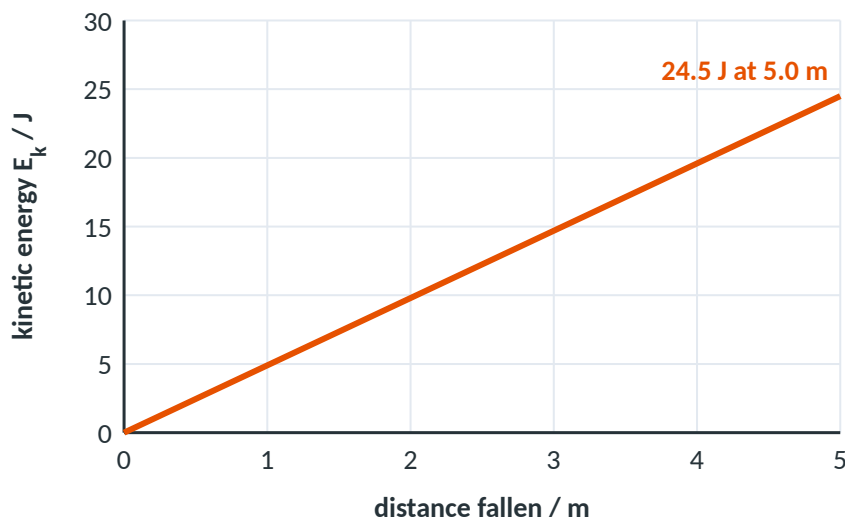
### Question 9

A spring ( $k = 200 \text{ N m}^{-1}$ ) is compressed by 0.050 m and launches a 0.020 kg toy straight up. Ignoring losses, how high does the toy rise?

- (A) 0.64 m
- (B) 1.27 m
- (C) 2.55 m
- (D) 51.0 m

### Question 10

A 0.50 kg ball is dropped. The graph shows its kinetic energy against the distance fallen. What does the gradient of the graph represent?



- (A) The ball's speed,  $4.9 \text{ m s}^{-1}$
- (B) The ball's mass, 0.50 kg
- (C) The acceleration of free fall,  $9.81 \text{ m s}^{-2}$
- (D) The ball's weight, 4.9 N